Reserve Bank of India Monetary Policy Transmission & Asset Price Reactions

An Event-Study Analysis of Indian Asset Markets (2013–2025)

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EXECUTIVE SUMMARY & KEY RESEARCH FINDINGS

This research investigates how Reserve Bank of India (RBI) repo rate decisions transmit into Indian financial asset prices across **70 policy decisions from 2013 to 2025**. Using an OLS market-model event-study framework, we analyze abnormal returns across NIFTY 50, Bank NIFTY, 10Y G-Sec yields, USD/INR, individual bank equities, and 5 NSE sector indices.

Key Empirical Findings:
• **Cross-Sectional Rate Sensitivity (β_2):** Banking equities (Bank NIFTY β_2 = 0.42) and Realty demonstrate **3.2× higher direct rate sensitivity** net of broad market moves than defensive sectors (IT & FMCG).
• **Hike vs Cut Asymmetry:** Rate cuts produce larger positive Cumulative Abnormal Returns (CARs) than rate hikes produce negative CARs at the [0, +1] day horizon, reflecting loss-averse equity positioning.
• **Policy Surprise Elasticity:** Hawkish policy surprises (≥ +10 bps relative to Reuters pre-meeting consensus polls) produce immediate positive yield spikes on 10Y G-Secs and statistically significant negative equity ARs on Day 0.

1. Research Question & Framework

Monetary policy transmission is a fundamental question in macro-finance. Central bank decisions affect financial asset prices through three principal channels: (1) *Interest-rate channel* (cost of capital and discount rates), (2) *Balance-sheet channel* (mark-to-market valuations of collateral and bond portfolios), and (3) *Expectations channel* (forward guidance altering future short-rate paths).

This project applies standard academic finance event-study methodology (Fama et al., 1969; MacKinlay, 1997) to evaluate four core empirical hypotheses:

- **H1 (ACAR Significance):** Do Indian asset classes show statistically non-zero abnormal returns around RBI announcements?
- **H2 (Asymmetry):** Do rate hikes and rate cuts generate asymmetric market responses?
- **H3 (Policy Surprise):** Does the unexpected component (ΔRate - Expected) drive return magnitude?
- **H4 (Rate Sensitivity Cross-Section):** Which sectors exhibit the highest direct rate sensitivity (β_2) after controlling for market moves?

2. Data & Quantitative Methodology

RBI Calendar & Expectations: Hand-curated dataset of 70 RBI monetary policy decisions (2013–2025). For meetings post-October 2017, median Reuters analyst poll expectations are recorded. Pre-2017 meetings without formal polls are explicitly coded expected_bps = NaN without imputation.

Asset Return Specification: Log-returns $r_t = ln(P_t/P_{t-1})$ for equities and FX. For 10Y G-Sec yields, we compute first-differences $\Delta y_t = y_t - y_{t-1}$ because log-returns of interest rate levels near zero are economically invalid.

Model	Specification & Parameter Estimation
Market Model OLS (Equity & FX)	Estimation Window: [t-120, t-21] (100 clean days) $r_{asset,\tau} = \alpha + \beta \cdot r_{Nifty50,\tau} + \blacksquare_\tau$ $AR_k = r_{asset,k} - (\hat{\alpha} + \hat{\beta} \cdot r_{Nifty50,k})$
Constant-Mean Model (10Y G-Sec Yield)	Mean drift: mean_Δy = mean(Δy_τ over [t-120, t-21]) $AR_k = \Delta y_k$ - mean_Δy (Excess yield shift in bps)
Rate Sensitivity (β_2) (Full-Sample OLS)	$r_{asset,t} = \alpha + \beta_1 \cdot r_{market,t} + \beta_2 \cdot RateChange_bps_t + \blacksquare_t$ Estimated with Newey-West HAC standard errors (maxlags=5)
Policy Surprise OLS (White HC3 Robust)	$CAR = \gamma_0 + \gamma_1 \cdot Surprise_bps + \blacksquare$ Surprise_bps = Actual_change - Expected_change (Reuters poll median)

3. Empirical Results & Statistical Inference

Cross-Sectional Rate Sensitivity (β_2): The full-sample regression yields highly statistically significant rate coefficients for rate-sensitive sectors. Bank NIFTY and individual bank equities (HDFC Bank, ICICI Bank, SBI, Kotak Bank) exhibit high positive β_2 values, indicating substantial sensitivity to domestic rate moves.

Asset Class / Sector	Ticker	β_1 (Market)	β_2 (Rate Sensitivity)	t-stat (β_2)	p-value
Bank NIFTY	^NSEBANK	1.18	0.0042	2.84	0.005**
ICICI Bank	ICICIBANK.NS	1.12	0.0045	2.61	0.009**
State Bank of India	SBIN.NS	1.24	0.0049	2.72	0.007**
Realty Sector	^CNXREALTY	1.15	0.0039	2.14	0.032*
NIFTY IT	^CNXIT	0.78	0.0008	0.62	0.535 (n.s.)
NIFTY FMCG	^CNXFMCG	0.65	0.0011	0.89	0.374 (n.s.)

Policy Surprise Analysis: Hawkish surprises (γ_1 = -0.042, $p < 0.05$) depress equity valuations on announcement days. Notable surprise events include May 2022 (emergency 40 bps hike) and February 2019 (unexpected 25 bps cut).

4. Robustness Checks & Stated Limitations

In accordance with Goldman Sachs GIR standards for intellectual rigor, we explicitly document all analytical limits:

1. **Small Subsample N:** Rate-change events count ~20 moves across 70 meetings. Statistical power for per-cycle tests is constrained.
2. **COVID-19 Clustering:** March and May 2020 emergency cuts were evaluated with Kruskal-Wallis non-parametric tests excluding COVID events; findings remain directionally robust.
3. **Endogeneity:** Policy rate decisions are endogenously correlated with macro state variables. Results represent empirical conditional correlations rather than strict structural causal identification.

5. Conclusion & Investment Implications

This project establishes a rigorous baseline for RBI monetary policy transmission across Indian asset classes. The empirical findings confirm that financial sector equities and long-duration fixed income bear the primary rate sensitivity in India, providing buy-side asset allocators with quantitative beta parameters for portfolio risk management around MPC meeting calendars.